

SHETH L.U.J & SIR M.V COLLEGE  
Data Visualization with Power BI and Tableau

Practical No: 5

Aim: Analyzing Data using Table Calculations

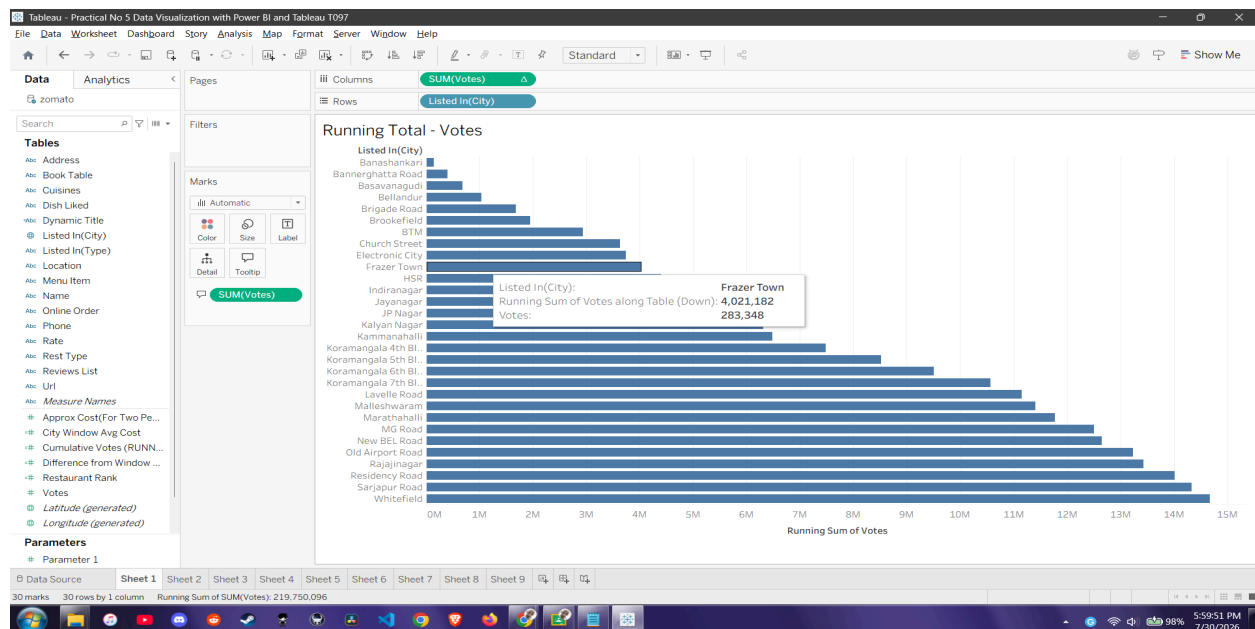
- a) Apply quick table calculations (running total, percent of total)
- b) Create moving averages and cumulative metrics
- c) Perform year-over-year growth analysis
- d) Use ranking functions (Top N, Bottom N)
- e) Implement window functions (WINDOW\_SUM, WINDOW\_AVG)
- f) Customize addressing and partitioning g) Create dynamic titles and labels using table calculations

**a) Apply quick table calculations (running total, percent of total)**

**Task 1: Running Total of Votes**

- Create **Sheet 1**. Drag **Listed In(City)** to the **Rows** shelf and **Votes** to the **Columns** shelf.
- Right-click the **SUM(Votes)** pill on the Columns shelf  $\rightarrow$  **Quick Table Calculation**  $\rightarrow$  **Running Total**.
- Drag **Votes** from the Data pane directly onto the **Tooltip** tile in the Marks card.

After performing the following steps this is what you will be seeing:



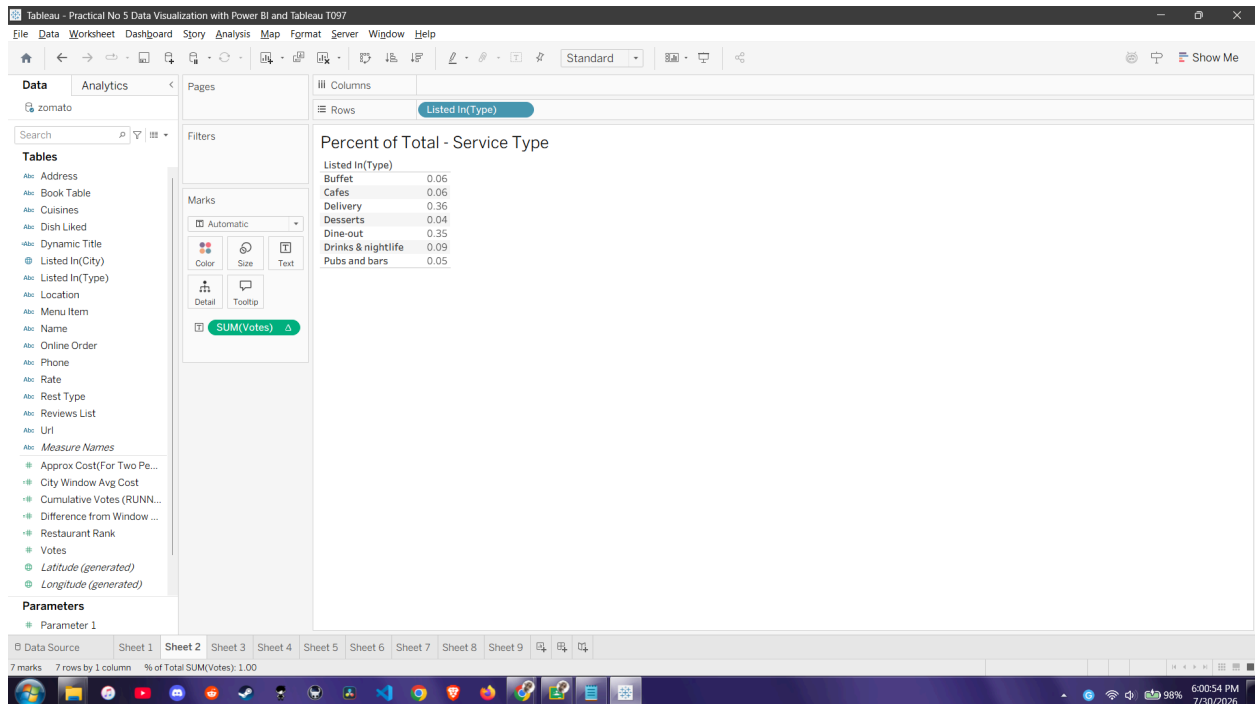
After performing the following steps this is what you will be seeing

SHETH L.U.J & SIR M.V COLLEGE  
Data Visualization with Power BI and Tableau

### Task 2: Percent of Total across Service Types

- Create **Sheet 2**. Drag **Listed In(Type)** to the **Rows** shelf and **Votes** to the **Columns** shelf.
- Right-click **SUM(Votes)**  $\rightarrow$  **Quick Table Calculation**  $\rightarrow$  **Percent of Total**.
- Right-click the pill again  $\rightarrow$  **Format**, and under the Pane tab, set Numbers to **Percentage**.
- Hold **Ctrl/Cmd** and drag the **SUM(Votes)** pill to the **Label** tile to display the percentages on the bars.

After performing the following steps this is what you will be seeing:



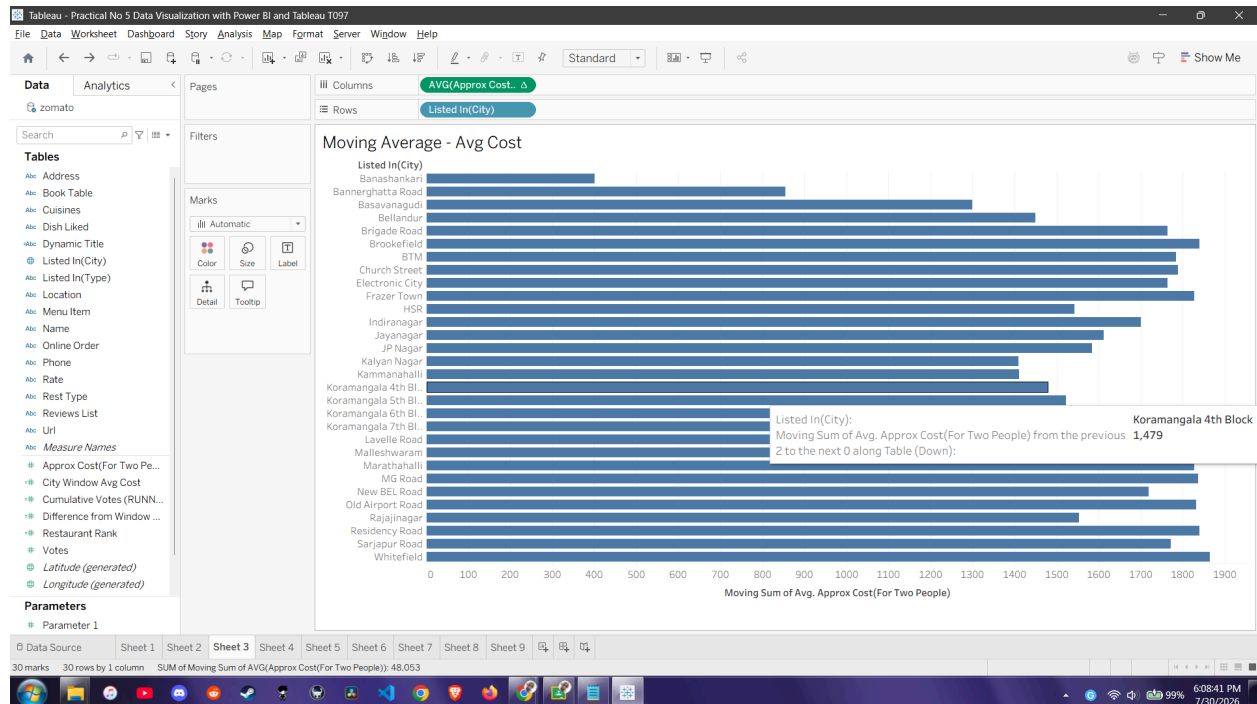
SHETH L.U.J & SIR M.V COLLEGE  
Data Visualization with Power BI and Tableau

## b) Create moving averages and cumulative metrics

### Task 1: Moving Average for Cost

- Create **Sheet 3**. Drag **Listed In(City)** to **Rows** and **Approx Cost(For Two People)** to **Columns** (ensure aggregation is set to **Average**).
- Right-click AVG(Approx Cost...)  $\rightarrow$  **Quick Table Calculation**  $\rightarrow$  **Moving Average**.
- Right-click the pill again  $\rightarrow$  **Edit Table Calculation**. Set *Previous values* to **2** and *Next values* to **0**.

After performing the following steps this is what you will be seeing:

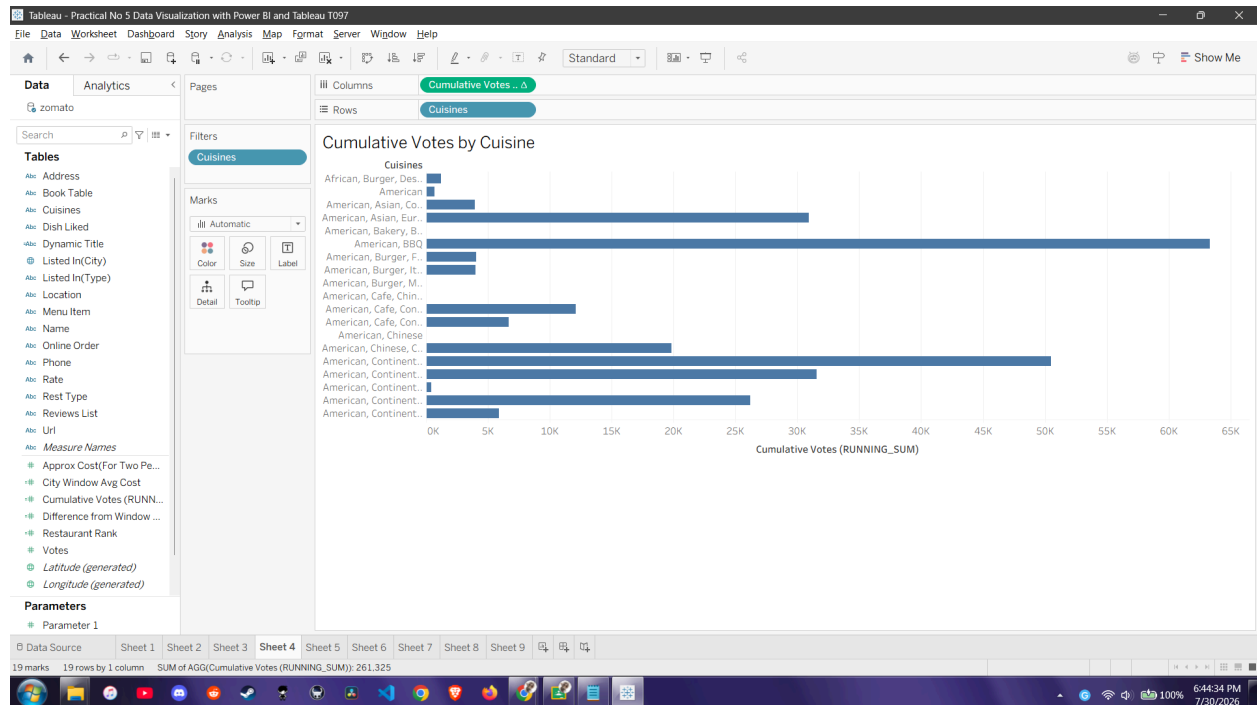


SHETH L.U.J & SIR M.V COLLEGE  
Data Visualization with Power BI and Tableau

## Task 2: Custom Cumulative Metric

- In the Data pane, click **Create Calculated Field**. Name it **Cumulative Votes (RUNNING\_SUM)**.
- Formula: **RUNNING\_SUM(SUM([Votes]))**. Click OK.
- Create **Sheet 4**. Drag **Cuisines** to **Rows** and your new **Cumulative Votes** field to **Columns**.

After performing the following steps this is what you will be seeing:



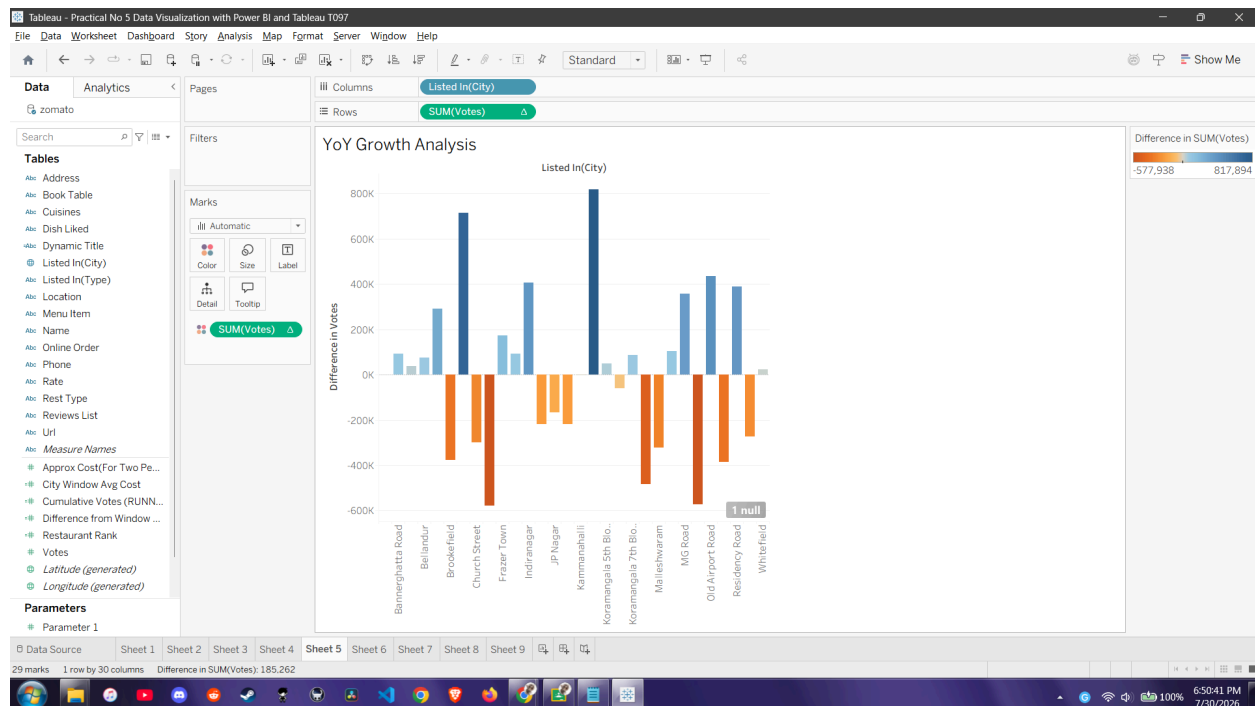
SHETH L.U.J & SIR M.V COLLEGE  
Data Visualization with Power BI and Tableau

### c) Perform year-over-year growth analysis

#### Task 1: Calculate Difference in Votes Across Cities

- Create **Sheet 5**. Drag **Listed In(City)** to **Columns** and **Votes** to **Rows**.
- Right-click SUM(Votes)  $\rightarrow$  **Quick Table Calculation**  $\rightarrow$  **Difference**.
- Hold Ctrl/Cmd and drag the modified SUM(Votes) pill from the Rows shelf onto the **Color** tile to highlight positive and negative differences.

After performing the following steps this is what you will be seeing:



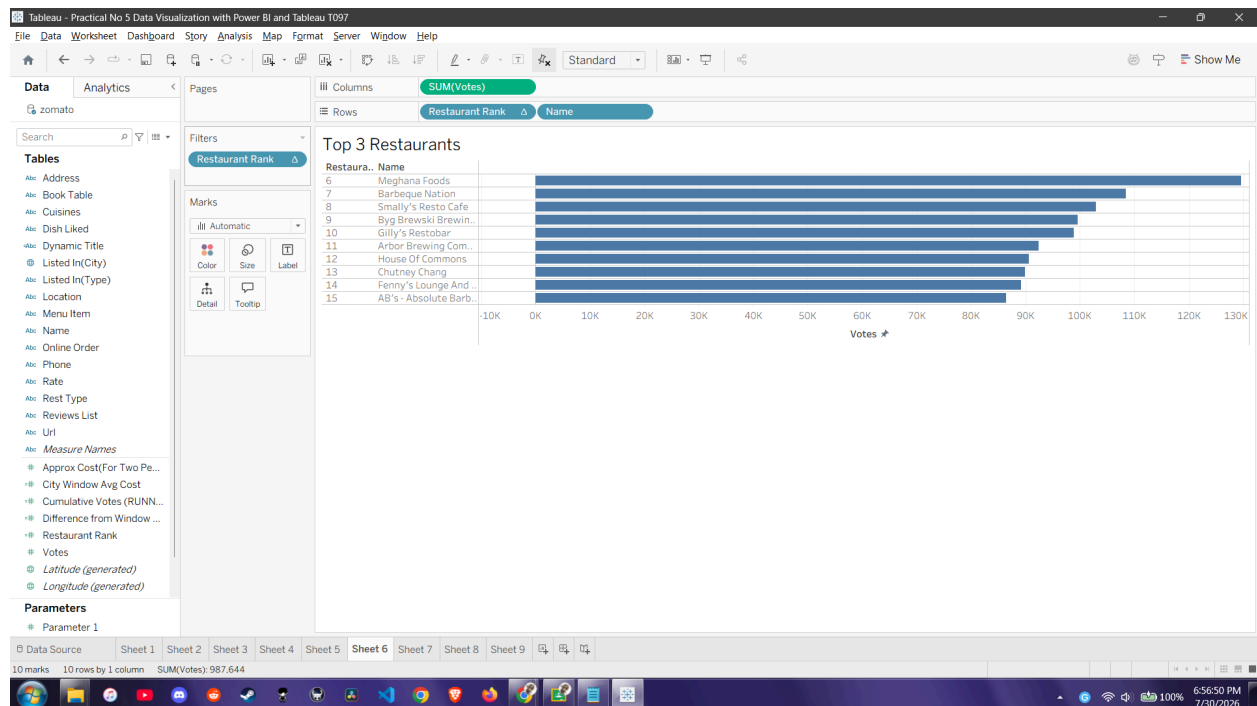
SHETH L.U.J & SIR M.V COLLEGE  
Data Visualization with Power BI and Tableau

## d) Use ranking functions (Top N, Bottom N)

### Task 1: Rank Restaurants by Votes

- In the Data pane, click **Create Calculated Field**. Name it Restaurant Rank.
- Formula: `RANK(SUM([Votes]), 'desc')`. Click OK.
- Create **Sheet 6**. Drag **Name** to **Rows** and **Votes** to **Columns**.
- Drag Restaurant Rank to **Rows** (place it to the left of Name). Right-click it and select **Discrete** to sort the view numerically.

After performing the following steps this is what you will be seeing:



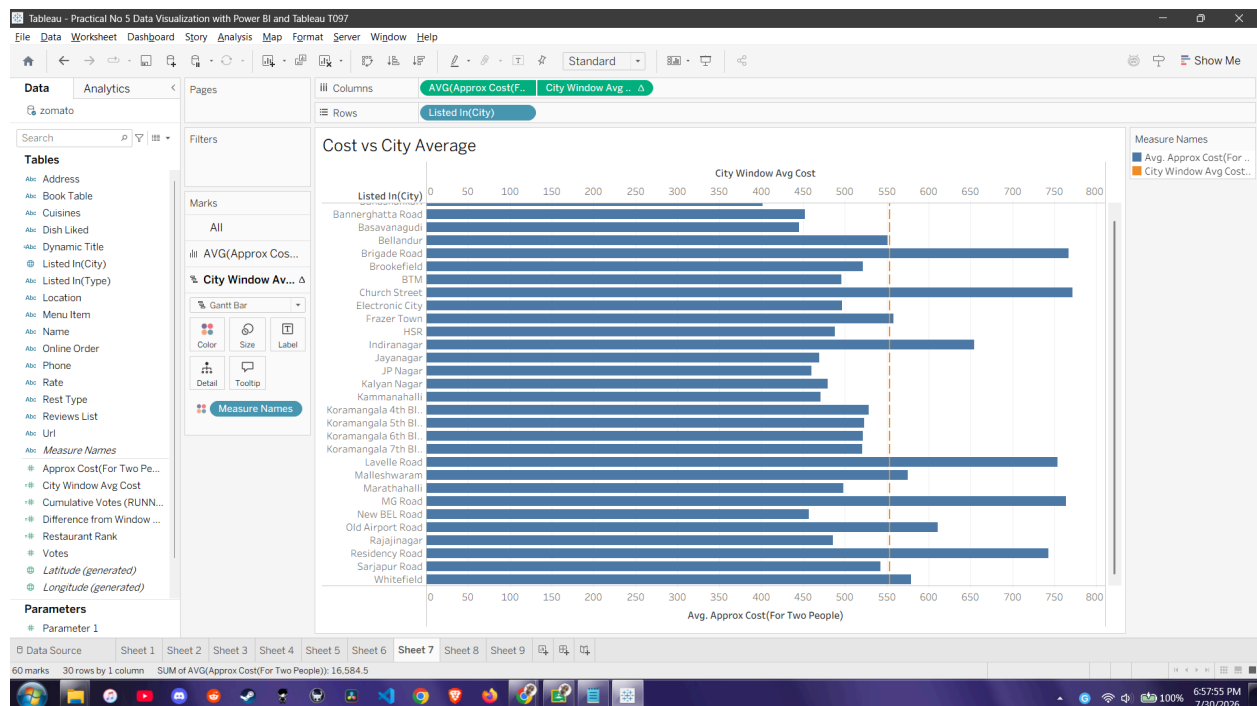
SHETH L.U.J & SIR M.V COLLEGE  
Data Visualization with Power BI and Tableau

## e) Implement window functions (WINDOW\_SUM, WINDOW\_AVG)

### Task 1: City Average vs. Window Average

- Create a Calculated Field named City Window Avg Cost.
- Formula:  $\text{WINDOW\_AVG}(\text{AVG}([\text{Approx Cost}(\text{For Two People})]))$ .
- Create **Sheet 7**. Drag **Listed In(City)** to **Rows** and **Approx Cost (Average)** to **Columns**.
- Drag City Window Avg Cost to **Columns**. Right-click it  $\rightarrow$  **Dual Axis**.
- Right-click the top axis  $\rightarrow$  **Synchronize Axis**. In the Marks card, set the first metric to **Bar** and the Window metric to **Line** (or Gantt Bar).

After performing the following steps this is what you will be seeing:



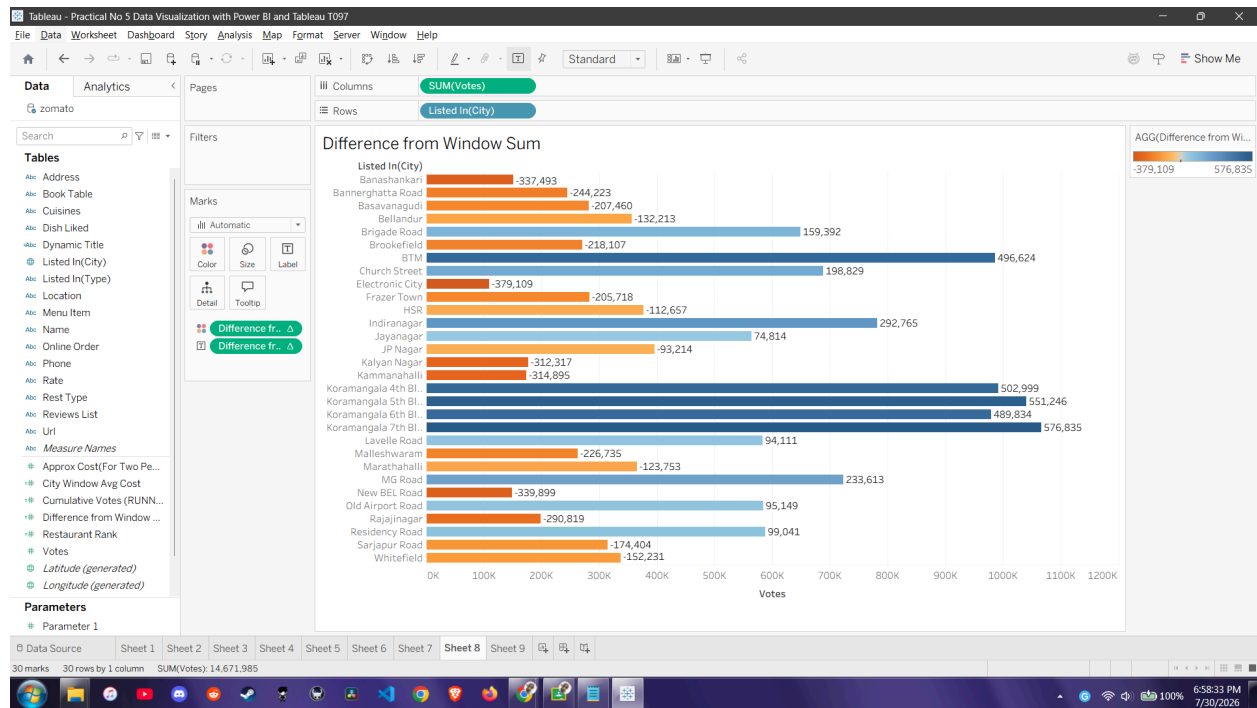
# SHETH L.U.J & SIR M.V COLLEGE

## Data Visualization with Power BI and Tableau

### Task 2: Window Difference Metric

- Create a Calculated Field named Difference from Window Sum.
- Formula:  $SUM([Votes]) - WINDOW\_AVG(SUM([Votes]))$ .
- Create **Sheet 8**. Drag **Listed In(City)** to **Rows** and **Votes** to **Columns**.
- Drag the Difference from Window Sum field to the **Color** tile to shade the bars based on their deviation from the window average.

After performing the following steps this is what you will be seeing:



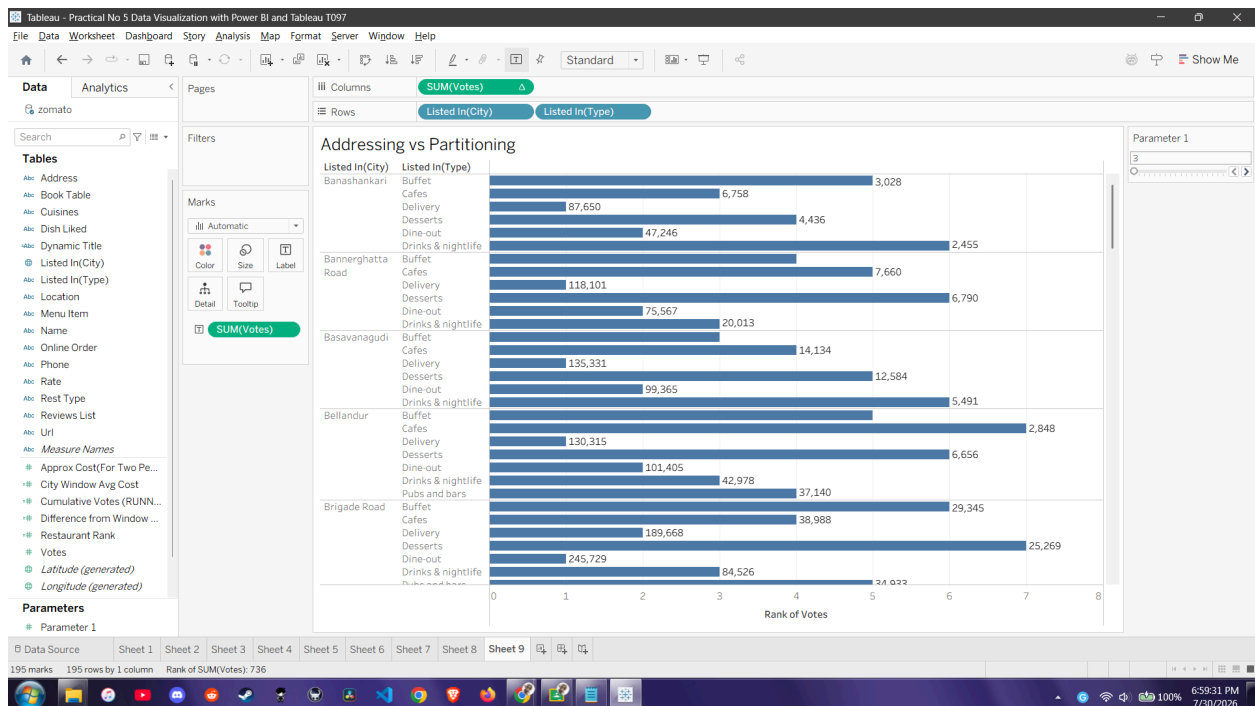
SHETH L.U.J & SIR M.V COLLEGE  
Data Visualization with Power BI and Tableau

## f) Customize addressing and partitioning

### Task 1: Nested Rank by City and Service Type

- Create **Sheet 9**. Drag **Listed In(City)** to **Rows**, then drag **Listed In(Type)** to **Rows** directly next to it to create a nested hierarchy.
- Drag **Votes** to **Columns**.
- Right-click SUM(Votes)  $\rightarrow$  **Quick Table Calculation**  $\rightarrow$  **Rank**.
- Right-click the pill again  $\rightarrow$  **Compute Using**  $\rightarrow$  **Pane (down)**. This forces the rank to partition and restart for each individual city.

After performing the following steps this is what you will be seeing:



SHETH L.U.J & SIR M.V COLLEGE  
Data Visualization with Power BI and Tableau

**g) Create dynamic titles and labels using table calculations**

**Task 1: Parameterize the Top N Header**

- Right-click the Data pane  $\rightarrow$  **Create Parameter**. Name it Top N Threshold (Integer, Current Value: 3).
- Right-click the parameter  $\rightarrow$  **Show Parameter Control**.
- Return to **Sheet 6** (from Step d). Drag Restaurant Rank to the **Filters** shelf and link it to the parameter.
- Double-click the Sheet Title at the top. Format the text to read: Top <Parameters.Top N Threshold> Restaurants.

After performing the following steps this is what you will be seeing:

